Identification of Bioactive Structures of Natural Compounds of Estrogen Receptor-Positive Breast Cancer Treatment

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Abstract

Research conducted for monitoring the activity of mugwort that is used in Korean traditional medicine for polycystic ovary syndrome(PCOS) which is accompanied by menopausal symptoms is not sufficient. Therefore, in order to determine an effective method, reviewing similar research conducted for breast cancer is necessary since it is the most prolific cancer amongst women with a larger breadth of research conducted in the area of finding natural remedies which this paper could draw from. As 67 - 80% of BrCa in women are ER-positive, and especially for ERα, as it mainly contributes to breast cancer cell growth, this paper focuses on identifying concrete structures of natural compounds that can actively treat ER-positive BrCa using 3D pharmacophore/3D QSAR models. The models are generated based on the known data sets of ligands that are bioactive against ERα with their pIC50 values. The generated models indicated that inhibitors need hydrogen-bond donators/acceptors, hydrophobic features, positive ionizable features, and ring features that can interact with steric force and electrostatic force fields. The most potent compounds are chosen as Hits and among them, through filtration, the most Hits are selected. The *in vivo/ in vitro* experiment methods are designed based on typically adapting Nezrina et al(2022)'s experiments. In the case of *in vitro* experiments, ER+ breast cancer cell lines are chosen, and for the control group, ER- breast cancer cell line and normal cell line are chosen, and for pIC50 values and ChIP assay results, and cell phases screening are chosen for observation methods. For *in vivo* experiment, observing the tissues of rats with mammary tumors induced by MNU is chosen. For further research on finding bioactive compounds in mugwort that can possibly treat PCOS, the 3D pharmacophore/3D QSAR models that would be generated should be using known data sets of ligands that are bioactive to ER and also that can induce its activity. From previous studies, eriodictyol and apigenin turned out to be estrogenic flavonoids, thus, these can be possible Hits chosen after the virtual screening is executed with models generated.

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1. Introduction

Breast cancer is the most common cancer among women cancers in the world. About 24.5% of women diagnosed with cancer worldwide had been diagnosed with breast cancer (BrCa) in 2020 (1). Most breast malignancies are adenocarcinoma, and they can be divided into noninvasive breast carcinomas, where abnormal cell proliferation is confined in the ducts and lobules, and invasive breast carcinoma which penetrates through the duct wall into the stroma (2). These BrCas can be again divided into various subtypes depending on the traditional classification systems regarding biological characteristics such as lymph node involvement, tumor size, and status of receptors in the cancer cells (3). Especially in BrCa cells, the level of estrogen receptors(ER), progesterone receptors (PR), and human epidermal growth receptor 2 (HER2) are used in determining the patient-tailored treatment strategies (3). Especially, the BrCas that have relatively high expression of estrogen receptors are luminal A breast cancer, Luminal B breast cancer (4-7). In the case of Liminal A breast cancer cells, the status of the cell is progesterone receptor-positive and HER2-negative, and it has a low level of the protein Ki-67, a protein that accelerates cancer growth($ER^+/PR^+/HER2^-$/low Ki-67)(42,66). Luminal B breast cancer is progesterone receptor-positive, HER2 positive, or a high level of Ki-67($ER^+/PR^+/HER2^+$ or ER^+/PR^+/high Ki-67), which indicates that the growth of cancer cells is faster than the previous cell type(3-6). As 67 - 80% of BrCa in women are ER-positive, endocrine therapy has been conducted on the patients as a complementary to surgery these days (8-10). Clinical parameters of ER status are also considered with the menopausal status since estrogens are produced mostly in the ovary for premenopausal women and in the aromatase for postmenopausal women (7).

Women who are premenopausal are treated with ovarian ablation, blocking the ovarian function. It can be done by physical operation removing the ovaries (oophorectomy) or treatment with radiation. Otherwise, drugs like goserelin (Zoladex) and leuprolide (Lupron), which serve as agonists of gonadotropin-releasing hormone (GnRH/LHRH) are used to suppress the ovary function (7). For those postmenopausal women, drugs target aromatase, the enzyme that turns androstenedione into E2 in the fat tissues.The aromatase inhibitors(AI) include anastrozole (Arimidex), letrozole (Femara), and exemestane (Aromasin) (13). However, these can also be used for premenopausal women to suppress ovarian function (15). Tamoxifen (Nolvadex), toremifene (Fareston), and raloxifene (Evista) are selective estrogen receptor modulators (SERMs) that compete with estrogen binding to the ERs. Fulvestrant (Faslodex) is another antiestrogen that binds to ER irreversibly and destruct the receptor (11). These drugs are usually used for premenopausal females; however, women who are postmenopausal are also can be treated with tamoxifen and fulvestrant (Faslodex) as they can block the ERs, especially for the fulvestrant as it is able to actually damage ER (12).

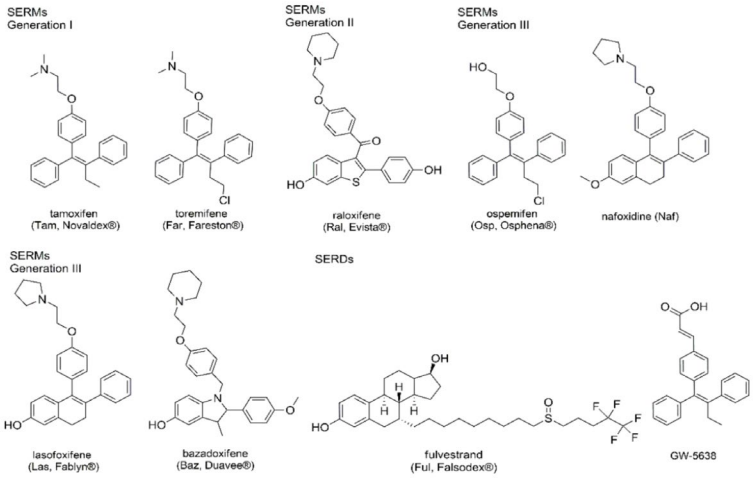


Figure.1. Structures of ERα affinity-selective agonists of FDA-approved SERMs and SERDs (22)

These current hormone therapies manifested several side effects and toxic effects that vary largely on the type of the treatment (14) with the association of resistance (18) from headaches to endometrial cancer and uterine sarcoma, vomiting, and breathing problems (19). Thus, a number of recent studies focus on investigating and evaluating different natural

compounds to assess their efficacy of medicinal properties in replacing the current therapies that can still target cancer cells with minimal or no side effects (18,19). In certain cases, natural compounds can be faster, inexpensive, and safer in the treatment of BrCa (16). For recent research investigated different natural compounds and evaluated their efficacy against several receptors or specific target domains and found diverse roles, such as inducing apoptosis or functioning as hormonal treatment (17).

This paper, therefore, will be focusing on finding concrete structures of natural compounds from previous studies that can actively treat ER-positive BrCa as alternatives to conventional chemical drugs. This research also aims to find an optimal method for finding bioactive compounds that can treat the targeted domain of the cells or receptors and suggest these methods for investigating other women's diseases' natural compounds, such as polycystic ovary syndrome(PCOS). To the best of the authors' knowledge, no significant study has been conducted for identifying the novel natural compounds that can treat PCOS. Therefore, this study will also suggest some of the natural compounds as candidates for future study using methods introduced in this paper.

2. Mechanism of ER-positive breast cancer cell growth and its clinical therapy

2.1. Estrogen Receptor's structure and interactions with the FDA-approved drugs

Estrogen signaling is a crucial regulator of BrCa, and thus ER signaling is the most attractive target for clinical treatment of ER-positive BrCa(21). ER receptors are included in the nuclear receptor family and consists of 5 types of distinct domains; A/B domain(transcriptional activation function (AF-1) at the N-terminal), C domain (DNA binding domain), D domain (nuclear localization, hinge domain), E/F domain (ligand-binding domain), activation function 2 at the C-terminal) (20).

There are two types of ERs, ERα, and ERβ. They engage in estrogen signaling and regulation of the transcription of cellular processes such as driving cell growth, proliferation, and differentiation (20,22). Both show high homology in the DNA and ligand-binding domains but differ in the transcriptional activation function-1(AF-1) domain. Due to this difference, ERα induces cell proliferation in BrCa cells, while ERβ opposes the action by regulating the expression of ERα-regulated genes. Thus, understanding the ERα-mediated drive pathway of BrCa is considered necessary(20).

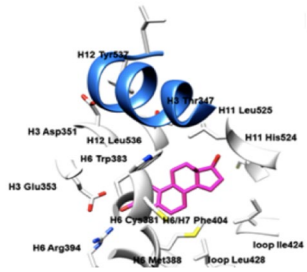


Figure.2. Key residues of ERα's active site (22)

The conventional FDA-approved SERMs and SERDs bind the ERα ligand-binding domain(Fig2.2), inducing its Helix 12 (H12) induced fitting(22). In the case of SERMs, they are non-steroid compounds that prevent ERα signaling of the genomic direct/indirect pathways. For steroid-based drugs, SERDs, allow the rapid downregulation and proteasomal degradation of ERα (22). This review paper will focus primarily on the studies identifying natural compounds that resemble SERMs.

2.2. E2 and ERα Signalling pathway

There are three pathways of E2 signaling pathways (Fig2.1.). First, E2 binding to ER and representing modulation of target gene transcription by binding to estrogen response elements (ERE) on DNA and representing interaction with coregulator proteins would be the classical role of the E2-ER signaling pathway(23). The second pathway would involve the cytoplasmic or membrane-localized ER to trigger cytoplasmic pathways and stimulates

membrane-associated receptors such as mGLUR(23). The last pathway does not include ER in the pathway, but rather E2 binds to the GPR30/GPER (G protein-coupled estrogen receptor). This pathway would lead to signal transduction engaged with cytoplasmic kinases (23).

In the case of ERα signaling pathways, direct interaction between ERα with the target gene promoters is mediated by the estrogen response element (ERE), while indirect interaction is mediated by the other DNA-bound transcription factors (25). This major mechanism of E2-ERα signaling can be divided into four types (Fig.2.2, 24). First, the E2-ERα complex directly binds to the ERE to the target genes and initiates the transcription. Second, E2-ER dimers interact with the transcriptional factors, which is an indirect way of regulating the target genes. Third, the E2-ERα complex initiates the cell signaling pathway of phosphorylation of transcription factors and activates gene regulation. Lastly, unlike the previous three ligand-dependent mechanisms, growth factor receptor (GFR) activation phosphorylates and dimerizes ERs and allows regulation of the target genes(23). The first to third mechanisms belong to the first and second pathways of E2 signaling. Understanding these signaling pathways is important since the increase in ERα-positive cells and elevated receptivity to E2 in ductal tissues are correlated to the higher risk of tumorigenesis (21).

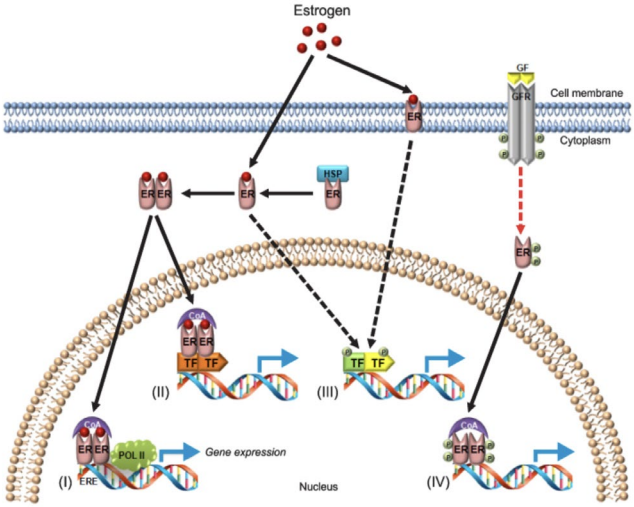


Figure.3. E2-ERα Signalling pathway (24)

2.3. Drive pathway of ER-positive breast cancer and current treatment

As there are so diverse kinds of signaling pathways and BrCa subtypes, this paper will be focusing only on the luminal A and B subtypes which are ER-positive BrCa subtypes and ER-engaged cancer drive pathways. The pathway of classical E2-ERα complex dimer binding to ERE with the coactivator for the execution of gene expression with RNApolymeraseII is the most simple pathway to understanding cancer cell growth. However, other signaling pathways engage more than one that leads to other signaling pathways.

The ERs in the cytoplasm and ERα that situates in the plasma membrane due to palmitoylation at cysteine 447 induce activation of MAP and PI3K/Akt pathways (26). They engage in the rapid action of E2 by the signal transduction pathways (27) which affects the proliferation, migration, and apoptotic inhibition of the BrCa(21,28). This paper will be dealing with the PI3K/Akt/mTOR signaling pathway that can be divided into two ways depending on the signal of Akt and mTOR (21), the signaling pathways of Src/Ras/MAPK/ERK, and the crosstalk of signaling pathways between E2-ERα and growth factors such as EGFR and IGF1R(21,4).

The loss of the controlling cell cycle is the main reason for cancer development and the role of cyclin proteins is known to be critical in endocrine and paracrine factor-induced mitogenesis in breast epithelial cells (21, 29). Cyclin proteins regulate the phase transition of

the G1 to S phase, and among these proteins, cyclin D1 is a target of the E2-ERα signaling pathway Akt (4,21). Cyclin D1 protein is encoded by the CCDNA1 gene and it is situated in the genomic region 11q13 of the chromosome where it is observed to be easily amplified in various human carcinomas including BrCa (21,30). Cylin D1 has two common polymorphisms; cyclin D1a and cyclin D1b, differing in carboxyl-terminal and existence of Thr-286 allowing only the cyclin D1a protein to be associated with ER activity (21). E2-ERα complex regulates cyclin D1 expression by involving transcription factors like AFT-2 and c-Jun (17,21), and thus, E2-induced upregulation of CCDN1 expression results in overactivation of CDK4 (21) and over phosphorylation of retinoblastoma protein (Rb protein) that are induced by cyclin D1a (21, 31). Overexpression of cyclin D1 in mammary epithelial cells leads to mammary carcinoma(21) failing in arresting mammary gland development (21). Also, Akt activates GATA4 (GATA binding protein 4), NFAT (nuclear factors of activated T cells), and NF-κB(nuclear factor-kappa B) transcription factors that induce cell survival, tumor-cell proliferation, and angiogenesis(32).

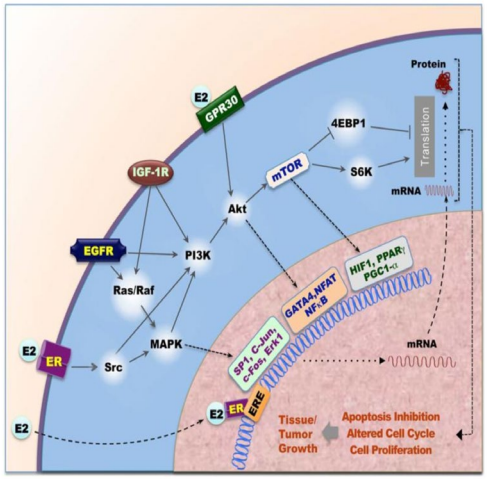


Figure.4. Drive pathway of ER-positive cancer (21)

In the case of mTOR signaling, it activates two enzymes, ribosomal protein S6 kinase beta-1(S6K) and eukaryotic translation initiation factor 4E-binding protein1 (4EBP1), which initiates translation and protein synthesis (21). Also, it activates transcription factors HIF1(hypoxia-inducible factor 1-alpha), PPAR-γ(peroxisome proliferator-activated receptor-gamma), and PGC1α (peroxisome proliferator-activated receptor-gamma coactivator) which attribute to ATP required for tumor growth (33).

Also, the rapid action of E2 stimulates the crosstalk of E2-ERα and various growth factor receptors like EGFR and IGF-IR(21). In this signal transduction, the effector molecules (PI3K/Akt, MAPK) are activated by the adaptor proteins (SHC-transforming protein1, Src kinase) (21). Src kinase activates the Ras/MAPK/ERK signaling pathway which induces BrCa cell proliferation (21) and also activates PI3K/Akt pathway that promotes BrCa cell growth(21). Shc proteins activate mitogen-activated protein kinases (MAPK) and PI3K/Akt signaling pathways (34, 35).

3. Methods used to analyze bioactive compounds against ER-positive breast cancer

3.1. Ligand-based drug design and structure-based drug design

Ligand-based drug design(LBDD) and structure-based drug design (SBDD) are the main methods used to develop hits and leads for drug candidates (20). LBDD focuses on designing new ligands that have more affinity against the proteins than pre-existing ligands (36). This method utilizes the preexisting data of the conformation of receptors or ligands by using pharmacophores, quantitative structure-activity relationship (QSAR), and machine learning techniques (22,36). In the case of SBDD, it proposes new ligands/ small molecules based on the structure of the binding pockets, and this includes pharmacophore, molecular docking, and molecular dynamics, (22, 36, 38). These two methods are chosen based on the available data of the know ligands and target proteins' structures for building virtual screenings and these are often used individually or together based on the studies (22, 36,39,40).

3.2. Data Preparation of ERα ligand-binding domain structure

Protein Data Bank (PDB) is used for the preparation of ERα's crystal structures. Studies selected co-crystallized structures of ERα with ligands (SERMs, SERDs, E2, partial agonists, etc.) bound to the binding domains (22, 36). The number of data selected varied among the studies, and this paper referred to the structure of ERα formed in Nezrina *et al.*(2022) since it covers the most diverse types of crystallized structure data of ligands binding to the wild-type ERα and mutated ERα.

The value of pIC50s (−log[IC50]) or pKi (−log[Ki]) is then gathered for each of the compounds which are used for building the training sets and test sets(22, 24, 36, 41). For the molecules in the training sets, they are classified as active or inactive depending on the pIC50 threshold value that is chosen in the individual studies. These compiled training sets and test sets have been used to train and evaluate the prediction ability of the optimal 3D pharmacophore models and the associated 3D QSAR models generated.

3.3. Generation of ERα 3D pharmacophore models and 3D QSAR models

Schrödinger's PHASE program or Discovery Studio (DS) v3.1's Hypogen algorithm is usually used to build the 3D pharmacophore models using four to five features; hydrogen bond donors, hydrogen bond acceptors, ring aromatic, and hydrophobic features or positively ionizable features (22). Also, 3D QSAR models are built by using the training sets with Schrödinger's PHASEprogram or Feature Mapping protocol and Create 3D Fingerprint protocol in DS. Through these, Hydrogen Bond Acceptor, Hydrogen Bond Donor, Positive Ionizable, Hydrophobic, and Ring Aromaticity features are mapped in the formation of the structures of ER, chemical traits, and force fields formed between the ligands and ER are gathered from training sets(22,36). Then the best 3D pharmacophore model and 3D QSAR model is chosen by validation of the models (22, 36).

From the hypotheses generated in the pharmacophore model, the ligands that are active and are having affinity to ERα show common features of the hydrogen-bond acceptor, hydrogen-bond donators, hydrophobic features, or aromatic ring structures, and also some with positively ionizable features (22,36). Also, from the combination of both models in Nezrina *et al*(2022), it turned out that the interaction between the ligand's hydrogen-bond donator and the region where negatively charged functional groups and hydrogen-bond acceptors are favored, forming negatively steric interactions, which recommends mixing of hydrogen-bond donators and hydrogen-bond accepting functional groups in order to form hydrogen bonding with H3 Glu353 and H6 Arg394. Another hydrogen-bond donator interacting with the region where a positive steric force field is formed and with the region where negatively charged functional groups and hydrogen-bond acceptors are favored implies that *para*-positioned hydrogen-bond acceptors and donator functional groups are needed to form hydrogen bonding with the H11 His524(22, 43-46). There are three hydrophobic features needed for the binders, and these interact with positive, and negative steric force fields, suggesting phenol hydroxyl groups to interconnect with five- to six-membered heterocyclic aliphatic bridge to interact with H6 Met388, H6-to-H7 loop residues Phe404, Ile424, and Leu428. These interactions with the region where negatively charged functional groups and hydrogen-bond acceptors are favored, also suggest that binders need a central phenyl ring to sterically interact with the H3 Thr347 side chain methyl group and inhibit the H3 Thr347-H11 Leu525-H12 Leu536 hydrophobic interaction formation. Interaction of the aromatic ring feature in the region where negative steric force field and negatively charged functional groups with hydrogen bond acceptors are favored shows the electrostatic interaction of an ethanolamine's oxygen atom and, an extension of phenyl group with the H3 Thr347′s side-chain -OH group. Lastly, the positively ionizable feature of the binder interacts with the region where a positive, negative steric force field is generated, and thus discerns the SERMs. This implies that the SERMs should form a hydrogen bond with H3 Asp351 with the hydrogen bond donators (e.g. positively charged nitrogen in

heterocyclic). The positive ionizable feature of the ERα binder also stabilizes the open form of the H12 open conformation and at the same maintains the minimal steric pressure on the H12.

3.4. Validation of 3D pharmacophore/3D QSAR model

Depending on the studies and programs used, the validation methods varied, thus this paper will only deal with common methods of validation used. Before evaluating the predictive ability of the models, the models are trained with the training sets(22, 36, 47,). Compiling training sets again varies depending on the studies. The most common way used is to group compounds from the know datasets, that have ligands that are bioactive against ERα and with their pIC50 values depending on the range of pIC50 values (22, 47). After training, the test sets are compiled with the compounds that are from the external datasets from the training sets' data sets with known pIC50 values and assess the models' predictive ability(22,36,47).

3.4.1. Fischer's randomization

Fischer's randomization is a method of validating the correlation between the structure of the compounds or ligands and their bioactivity(16, 22,36, 39, 47-53). A good model should be producing better RMSD and significant correlation values (48) compared to randomized sets. The experimental values of the training sets are randomly distributed to generate random pharmacophore models using the same criteria used to generate the original models in the previous steps. Then, a confidence level of 95% is selected and with these, spreadsheets are formed to score each hypotheses' significance using the formula [{1-(1+X)/Y}*100] where X refers to the sum of the hypotheses, while Y refers to the total sum of the initial and the random runs(36, 47). If the model that is generated is a good model, its total cost should be lower than any other hypothesis' cost values(47).

3.4.2. Test set prediction

The test set prediction is to generate the sets of compounds that are from the external data sets other than the data sets from the training sets(22, 47). The compiled test sets are put into the model and the predictive values of pIC50 are gathered. Then, with the experimental values of pIC50, the correlation coefficient value(R^2) is computed. The correlation coefficient value of the experimental values of the pIC50 versus to predictive (recalculated) pIC50 value should be close to 1 and also higher than the other randomly generated hypotheses(22, 36,47) generated by using the Fischer's randomization(22,36, 47, 48).

3.5. Virtual screening for Hit and docking analysis

A validated model is then used for virtual screening for the identification of Hits in various natural compound databases. Commonly used databases among the studies found in this paper are NCI(National Cancer Institute), ZINC TCM Database@Taiwan, Maybriddge, Asinex, and Cambridge (22, 36, 47, 49). In this section, this paper will only be reviewing the most reasonable method used by Nezrina *et al*(2022). It used the data from the NCI consisting of 4411 compounds and among them, 18 virtual Hits are chosen. Then through predicted pIC50 values, it categorized the compounds into ERα binders or antiproliferative agents against MCF-7. MDA-MB-231, and MRC-5 cancer cell lines. In this process of filtration, methods that are not introduced in Nezerina *et al*(2022) would be adapting Lipinski's rule of five; compounds should have a molecular weight less than 500, hydrogen donors less than 5, hydrogen acceptors less than 10, and an octanol/water partition coefficient value less than 5 (49). In addition, by using the ADMET and TOPKAT in DS, it is possible to calculate the ADMET(absorption, distribution, metabolism, and excretion) properties of the ligands, such as aqueous solubility, blood-brain barrier penetration, cytochrome P450 2D6 inhibition, plasma protein binding level, rodent carcinogenicity, and developmental toxicity potential(40). Through filtration, Brefeldin A (BFA, NCI code: NCI89671) is chosen as the most potent compound for the treatment by

showing the activity against the ERα value of IC50 of 8.34 μM. MCF-7 cell line value of IC50 9.01 μM, and selectivity against the MDA-MB-231 cell line. Still, BFA has less potency compared to E2 and raloxifene. This docking analysis can also be evaluated by calculating the docking score and fit value. In the case of docking score, it can be calculated with the formula of Dockscore = -(ligand/receptor interaction energy + ligand internal energy) (49). In the case of the fit value, it can be calculated

Observation of 3D pharmacophore/3D QSAR models describes that the binding of BFA acts against ERα as a partial agonist, and it induces the close conformation of H12. The main specific interactions generated between BFA and the receptor are the cyclopentane ring and the C7-OH group forming hydrogen bonds with H3 Glu353 and H6 Arg394, the C4-OH generating the electrostatic interactions with H3 Glu353. Other interactions would be the C9-to-C15 engaging in van der Waals interactions with H6 Met388 and H6-to-H7 loop residues Ile423 and Leu428 and C1 carbonyl group is considered as not satisfying any feature of models generated.

3.6. *In Vitro/ In Vivo* experiments

Studies that have found novel natural compounds against BrCa mostly did not develop the research to the *in vitro, in vivo* experiments. So again, this review paper will be suggesting the methods used in the Nezerina *et al*(2022).

For *in vitro* experiments, the BrCa cell lines that are ER+ can be used. Referring to Xiaofeng *et al*.(2017)'s classification of BrCa cell lines, about 25 cancer cell lines are ER+. In the cancer cell examination, by adapting Nezrina *et al*(2022)'s cancer cell experiments, MDA-MB-231(hormone-independent BrCa cell), MRC-5 (normal human lung tissue cell) were treated with the most potent Hits, and MCF-7(ER+ BrCa cell) treated with the E2 can be used for the control groups. For the experimental group, MCF-7 treated with the most potent Hits can be used. After applying those components to the cell lines, pIC50 values can be obtained. Also, using the ChIP(Chromatin immunoprecipitation) assay, the proliferative activity can be observed and compared among the cell line groups by confirming the existence of p53's expression which is a cancer-suppressing protein. In addition, observation of the cell cycle can be also checked by calculating and comparing the ratio of cells in the specific cell phases.

For *in vivo* experiments, rats can be treated with the 50mg/kg of methyl nitrosourea (MNU) that induces mammary tumors. Then, rats' mammary tissues in the control group that is not treated with the candidate natural remedies will be compared with those in the experimental groups that are treated with the compounds.

Further discussion

A number of studies had been already conducted to identify natural compounds that are bioactive and show effects on the receptor's activity, cancer cell lines, and rats against women's cancers (22, 36,47,48). Especially, research using the pharmacophore method and the 3D-QSAR method are easily found. However, for women's diseases such as polycystic ovary syndrome (PCOS), to the best of the authors' knowledge, no significant study has yet been conducted to explore structural data for generating 3-D pharmacophore models. Therefore, identifying natural compounds targeting women's diseases like PCOS is needed. For further research on finding the natural compounds for PCOS as a treatment, the writer suggests targeting two compounds Eriodictyol and Apigenin, whose structures are similar to the estrogen molecules (54-57). These compounds are from the *Artemisia Vulgari*s, also known as mugwort. Artemisia Vulgaris is known for its gynecological remedy for women who have an abnormal menopausal or irregular period of the menstrual cycle (amenorrhea, oligomenorrhea) (54-57). The studies that this paper reviewed were the studies targeting the natural compounds that can work as inhibitors, however, since PCOS is a disease accompanying menopause, the possible natural compound candidates should be the ones that can actually activate the estrogen receptors. Thus, when generating the 3D pharmacophore/3D QSAR models, the datasets of known should contain compounds that are E2-like such as eriodictyol and apigenin. However, since they only showed 5% of the estrogenic activity compared to estradiol(E2) in the previous research conducted(54-57), they can be considered weak estrogens that still compete with the E2 but can't be the most potent Hits(54-57). While these two natural compounds may expand the stimulating time length of estrogen receptors by competing with the E2 contributing to emmenogoic properties (54-57), still other potent Hits should be screened using the 3D pharmacophore method and the 3D-QSAR method.

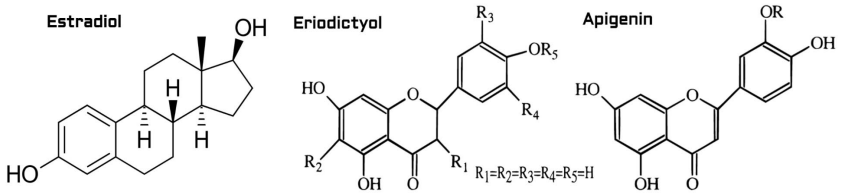


Figure.5. Structure of E2 and estrogenic flavonoids in Artemisia vulgaris L. (adapted from 57)

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